

DISTAL SENSING AND DISTANCE MEASUREMENT EMBODIMENTS

In various embodiments, the distal sensing region of the device is configured to determine distance between the device tip and an internal anatomical boundary or terminus. The distal sensing architecture may employ one or more ranging modalities selected based on cost, accuracy, manufacturability, or intended application environment.

In simplified embodiments, the distal sensing region includes a single time-of-flight ranging sensor configured to emit energy toward a target surface and determine distance based on measured return time. Optical time-of-flight sensors, infrared ranging modules, or similar compact ranging devices may be utilized to obtain scalar distance measurements sufficient for baseline profiling applications.

In certain embodiments, LiDAR-based time-of-flight sensing may be employed to improve robustness across variable surface properties, moisture conditions, or anatomical variability. Such LiDAR implementations may utilize single-point ranging, multi-zone ranging, or repeated sampling over time to improve measurement stability.

In alternative embodiments, the distal sensing region may incorporate an ultrasonic transducer configured to perform acoustic time-of-flight ranging. Ultrasonic sensing may provide improved performance under conditions where optical reflectivity varies or where acoustic coupling provides more reliable measurement behavior.

In advanced embodiments, the distal sensing region may acquire multiple measurements sequentially or simultaneously to improve accuracy through averaging, filtering, or sensor fusion. Multiple ranging samples may be used to compensate for motion artifacts, angular deviation, tissue variability, or transient environmental effects.

In certain configurations, the distal sensing region may additionally support partial scanning or imaging operations capable of generating a coarse geometric representation of the distal intraluminal space. Such reconstruction may be performed through repeated ranging measurements, motion-assisted sampling, or computational interpolation of sampled distance data.

A temperature sensor may optionally be incorporated near the distal sensing region to compensate for environmental or physiologic conditions affecting ranging accuracy, including acoustic propagation changes or optical performance variation.

The distal sensing architecture may further incorporate protective windows, acoustic coupling layers, hydrophobic coatings, or sealing structures configured to protect sensing components while maintaining measurement fidelity within intraluminal environments.

Distance measurements acquired by the distal sensing region may be processed locally within device electronics or transmitted to an external computational system for filtering, validation, and generation of derived distance indices suitable for user profiling, calibration, or integration into broader physiologic analysis frameworks.